

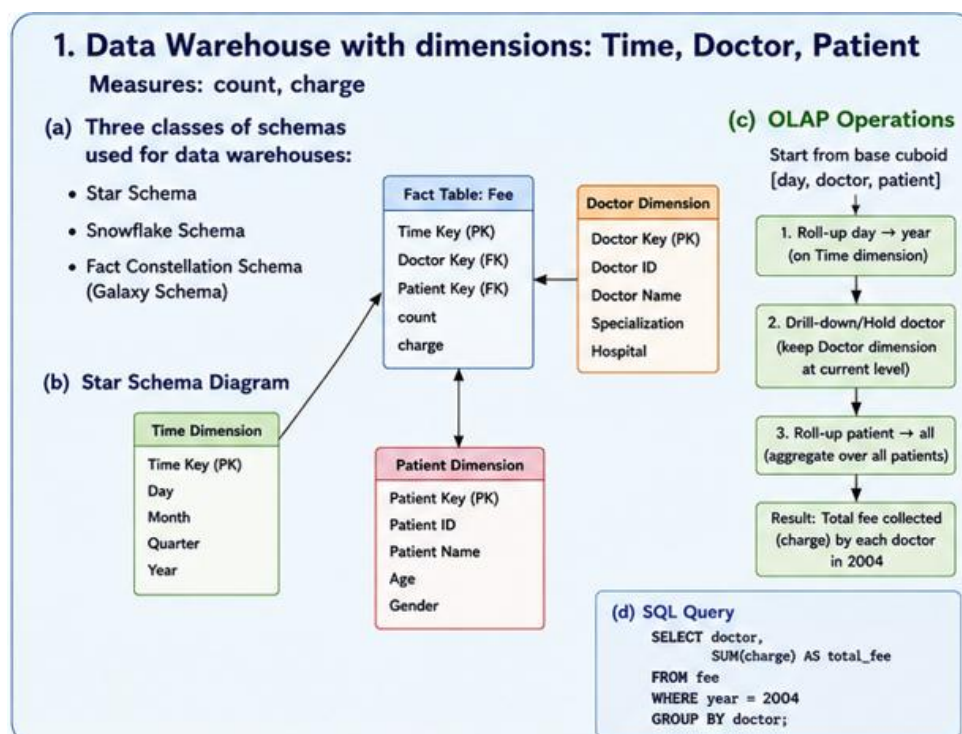
Activity-3

S. Sharon Surya

192324246

1. Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.

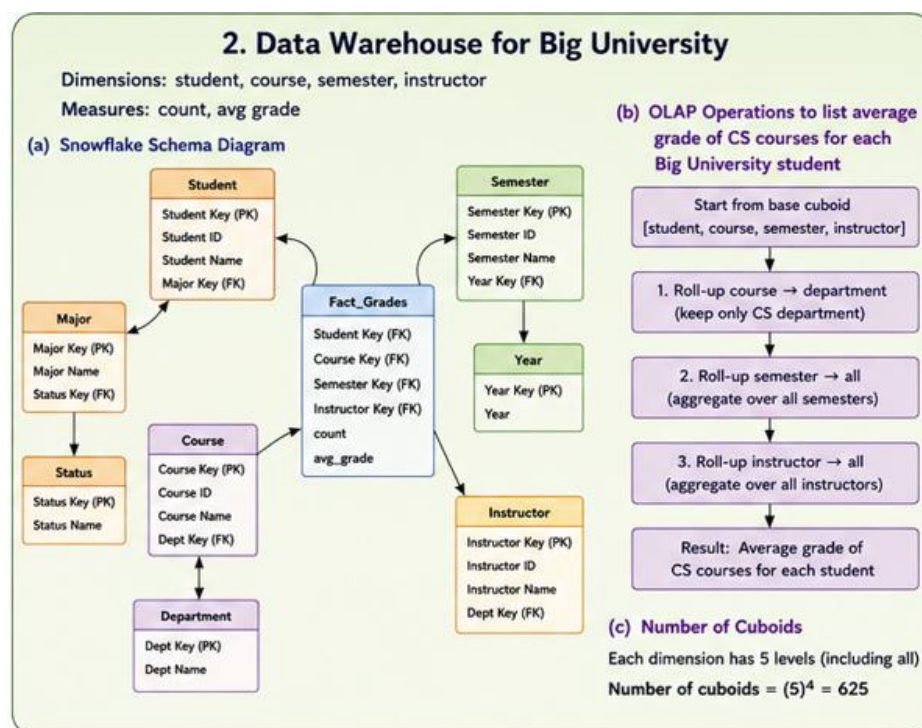
- Enumerate three classes of schemas that are popularly used for modeling data warehouses.
- Draw a schema diagram for the above data warehouse using one of the schema classes listed in (a).
- Starting with the base cuboid [day,doctor,patient], what specific OLAP operations should be performed in order to list the total fee collected by each doctor in 2004?
- To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge).



2. Suppose that a data warehouse for Big University consists of the following four

dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination.

1. (a) Draw a snowflake schema diagram for the data warehouse.
2. (b) Starting with the base cuboid [student, course, semester, instructor], what specific OLAP operations (e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses for each Big University student.



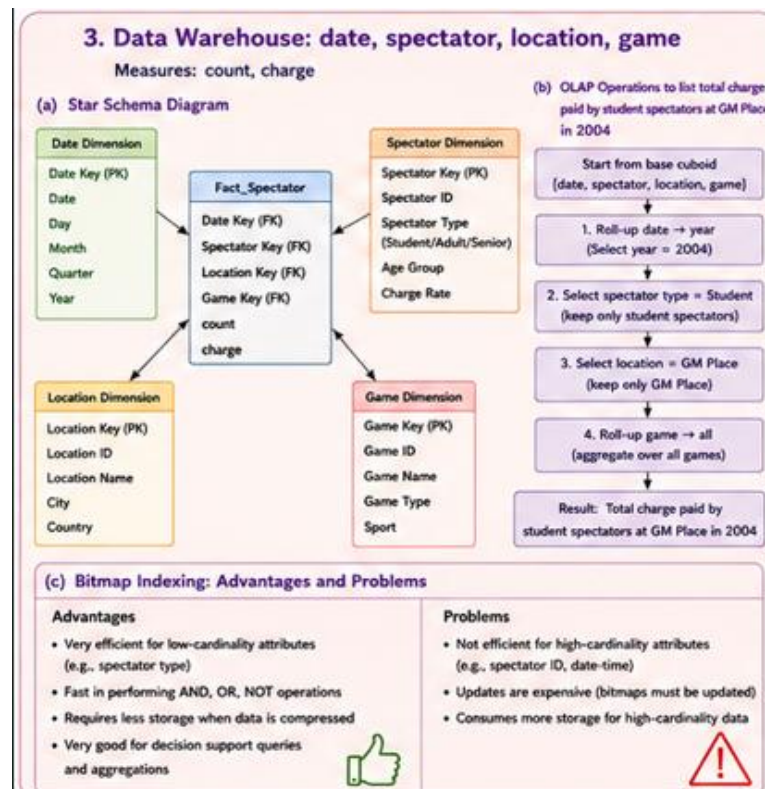
3. (c) If each dimension has five levels (including all), such as “student < major < status < university < all”, how many cuboids will this cube contain (including the base and apex cuboids)?

3. Suppose that a data warehouse consists of the following dimensions date, spectator, location, and game, and the two measures, count and charge, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults, or seniors, with each category having its own charge rate.

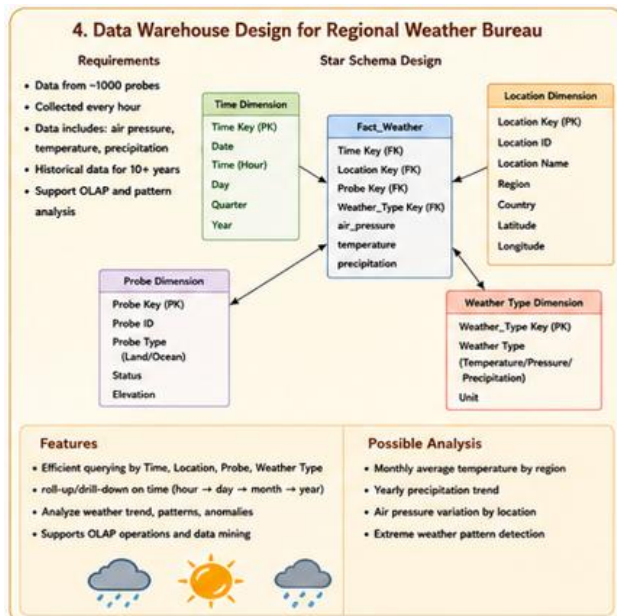
(a) Draw a star schema diagram for the data warehouse.

(b) Starting with the base cuboid [date, spectator, location, game], what specific OLAP operations should one perform in order to list the total charge paid by student spectators at GM Place in 2004?

(c) Bitmap indexing is useful in data warehousing. Taking this cube as an example, briefly discuss advantages and problems of using a bitmap index structure.



4. Design a data warehouse for a regional weather bureau. The weather bureau has about 1,000 probes, which are scattered throughout various land and ocean locations in the region to collect basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on-line analytical processing, and derive general weather patterns in multidimensional space.



5. A data cube, C , has n dimensions, and each dimension has exactly p distinct values in the base cuboid. Assume that there are no concept hierarchies associated with the dimensions.

- (a) What is the maximum number of cells possible in the base cuboid?
- (b) What is the minimum number of cells possible in the base cuboid?
- (c) What is the maximum number of cells possible (including both base cells and aggregate cells) in the data cube, C ?
- (d) What is the minimum number of cells possible in the data cube, C ?

